

How to Watch a Broadcast Like You Mean It

The TV picture follows the ball, and the ball is the least informative thing on the field. Where the cameras are, what they hide, what to read in five seconds before the snap, where to look after it, and how to keep score on your own guesses.

Second-and-8 at your own 25. The quarterback takes the snap in the shotgun and drops, and the camera stays with him: linemen grappling, his feet resetting, his head turning. Then the ball is gone, and for a second and a half you are looking at nothing in particular until the picture swings to the far sideline, where a receiver you'd forgotten about is catching it fifteen yards downfield with nobody near him. The replay from high behind the end zone shows what the live picture never did. Before the snap two safeties stood deep. At the snap one of them dropped toward the line, the other ran to the middle, and the cornerback on that side turned and sprinted deep, so the receiver who started deep and then stopped had five yards of grass to himself. All of that was decided in the first second, and none of it was on your screen. This chapter is about watching around that problem: knowing what the TV hides, using the seconds before the snap well, picking one thing to watch, and keeping score on your own guesses so that, by the end of this course, you can show you got better.

You'll be able to...

- Say what the main broadcast camera shows and hides, and which replay angle (tight, end zone, skycam) answers which question.
- Do a five-second pre-snap read of the score graphic: down, distance, field position, clock, score and timeouts, turned into "what does each side need here?"
- Pick one thing to watch on each snap, by phase (before the snap, the first second, the ball in the air) and by the kind of play you expect, instead of following the ball.
- Read the win-probability bar and Next Gen Stats numbers on screen as numbers with known meanings and known limits.
- Spot four starter tells (linemen's stance, the running back's depth and offset, receivers' splits, safety depth), say what each suggests, and say why each is only a clue.
- Keep a prediction log, score it, and compare it with your down-and-distance baseline and with what simple rules can do.

The broadcast is built to follow the ball

Start with what a broadcast is *for*. A network is telling a story to millions of people, most of whom want to know one thing: what happened to the ball. So the main camera, the one you watch for most of every snap, is a **game camera** high on one sideline near midfield, run by an operator whose job is to keep the ball, and the players closest to it, large and in the middle of your screen. That is a sensible choice for television. A picture wide enough to show all 22 players makes each of them small, and on a phone a tiny, sideline-to-sideline shot is hard to follow at all.

But for reading the game it is the wrong picture, because **the ball is the least informative thing on the field**. It is always in the same place at the snap. The information is elsewhere: in how deep the safeties stand, in which way the linemen step, in what the receivers do once they are twenty yards away. Those are exactly the things a ball-following camera crops out.

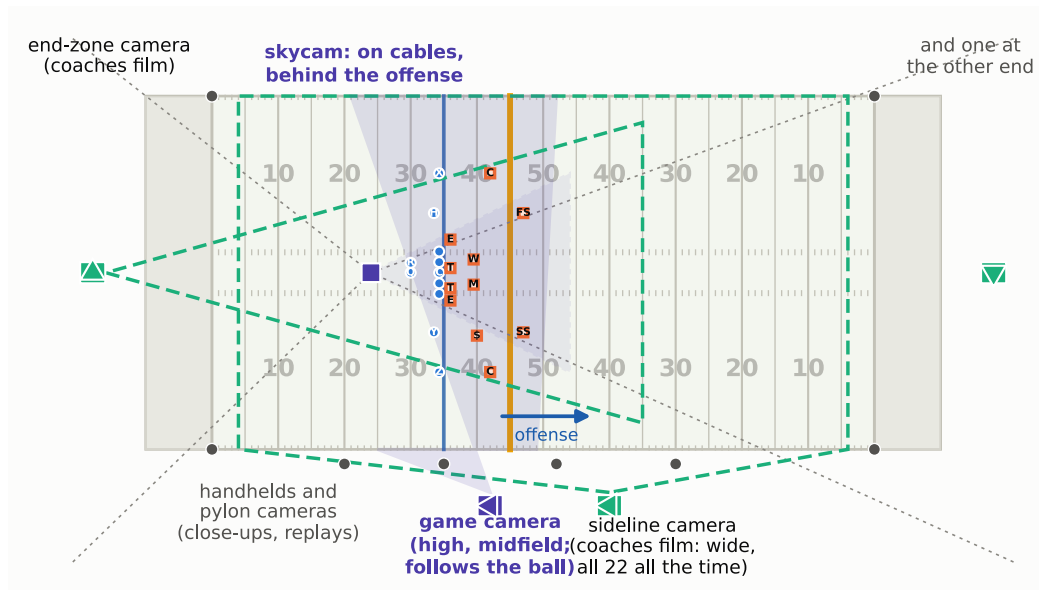


Figure 1: Where the cameras are (typical NFL setup; positions and counts vary by network and stadium). Purple: the broadcast's main angles. The game camera high on one sideline follows the ball with a tight frame; the skycam hangs on cables and flies behind the offense. Aqua: the coaches-film angles, a wide sideline camera and a high end-zone camera at each end, both zoomed out to keep all 22 players in view for the whole snap. Close-up handhelds and pylon cameras feed the replays.

The figure shows a typical setup. Networks bring many more cameras than this (big games bring dozens), but most of them feed close-ups and replays. Four angles matter for reading the game.

- The **game camera** you've already met. It sits high on one sideline and follows the ball.
- The **sideline camera** of the coaches' video crew sits in roughly the same place but is zoomed out wide enough to keep every player in the picture from the snap to the whistle. It never follows the ball, and it never cuts away.
- The **end-zone camera** sits high behind each end zone and looks straight down the field. From the side, the line is a pile of bodies. From behind, you can see the gaps between linemen, who blocks whom, and how deep each defender drops, which the sideline view flattens.
- The **skycam** hangs from four cables over the field and is usually flown a little behind and above the offense. Its view, from behind the quarterback looking downfield, is the one that looks most like a video game.

The skycam is older than it looks. It was invented by Garrett Brown, who also invented the Steadicam, and was tried out at an NFL preseason game in 1984. The XFL made it a main angle in 2001, and ESPN used it on its Sunday night NFL games from 2002.¹ Broadcasters still mostly use it for replays and some live snaps rather than as the main picture, because a behind-the-quarterback view makes depth hard to judge: you can't easily tell whether a receiver is eight yards downfield or fifteen.

Two kinds of video come out of these cameras. The **broadcast** is everything you see on TV: live pictures, replays, graphics. **Coaches film** is what teams shoot for their own study: every snap from the high sideline and end-zone angles, cut from just before the snap to the whistle, with no close-ups, no replays and no graphics. The wide sideline angle is what people mean by **All-22**: video in which all 22 players are visible for the whole play. When analysts and coaches "watch the tape," this is the tape.

i Madden vs. real life

Madden's default camera hangs high behind the offense and shows you most of the field: both safeties, every receiver, the whole defense lining up. That's essentially a skycam with the zoom pulled back, an All-22 view you get for free on every snap. A TV broadcast is not that. For most of a live play you see perhaps half the players, and almost never the deep safeties after the snap. If you've learned football from the game, part of what you know depends on a view the broadcast doesn't give you. You can get it back, partly from replays and fully from All-22 film.

¹Wikipedia, "Skycam" (invented by Garrett Brown; trial at a 1984 49ers–Chargers preseason game; the XFL's main angle in 2001; ESPN's NFL preseason use in 2001 and Sunday-night use from 2002): <https://en.wikipedia.org/wiki/Skycam> ; Sports Broadcasting Hall of Fame, "Garrett Brown": <https://www.sportsbroadcastinghalloffame.org/inductees/garrett-brown/>

What the broadcast frame hides

Here is the opening play, drawn from above the way the All-22 camera would see it. The shaded area is what a typical broadcast frame would show at each moment; players outside it are faded. The play is simulated (built in the same coordinates as the NFL's real player-tracking data, which later chapters use), and the TV frame is a simple model of a typical game camera, not a measurement of any network's coverage. But the pattern holds on almost every pass play.

Reading the field diagrams. The offense moves up the page. The solid line across the field is the line of scrimmage, and the thick yellow-orange line is the line to gain (eight yards ahead here). Offensive players are blue circles: **Q** quarterback, **R** running back, **C** center (the other four linemen are unlabeled), and pass-catchers **X**, **Z** (the two wide receivers), **H** (the slot receiver) and **Y** (the tight end; here he lines up away from the line, like a slot receiver). Defenders are orange squares: **E** end, **T** tackle, **M** and **W** linebackers, **C** cornerback, **\$** the slot cornerback (or nickel back), **FS** and **SS** the safeties. The full labeling system is in Formation Rules and the Language of Alignment.

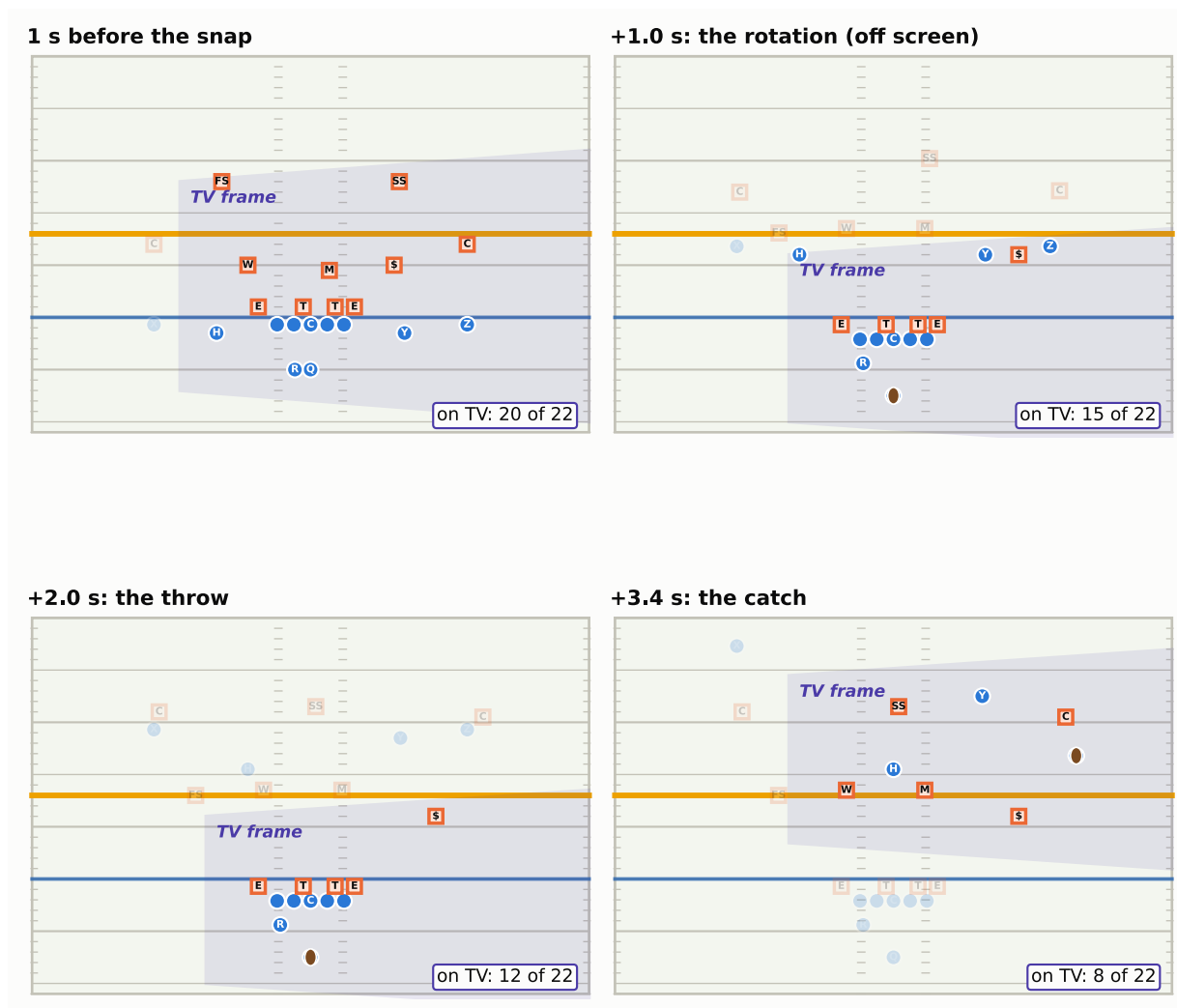


Figure 2: The same snap two ways at once: everything drawn is what All-22 film shows; the purple area is roughly what the broadcast shows (simulated play; the TV frame is a typical game-camera framing, camera on the left sideline). Before the snap the TV frame holds most of the formation. A second after the snap it has tightened on the quarterback, and the safeties' rotation, the cornerbacks bailing deep and three of the four receivers are off screen. The picture finds Z only as the ball arrives.

Walk through it.

Before the snap, the broadcast frame is generous. You can usually see the whole offensive formation except a receiver near your own sideline, all the linemen and linebackers, and, on most modern broadcasts, the safeties standing deep. This is the moment the TV gives you the most information, and it's the reason the next three sections are about using it.

At the snap the frame tightens on the quarterback. The camera has to: the play might be a handoff, a scramble, a short throw, and the operator needs the ball large and centered. So for the next second or two you are watching the least decisive part of a

pass play, the protection and the quarterback's feet, while the decisive part is happening off screen: the safeties choosing their shell (here they *rotate*, one dropping down and one running to the middle, a disguise that a whole later chapter is about), the cornerbacks turning to run, and the receivers getting into their routes. In the model, only about half of the 22 players are on screen at that moment.

When the ball is thrown, the camera chases it, and you arrive at the receiver at the same moment the ball does. That is why a deep completion on TV often looks like magic: the receiver appears from nowhere, wide open, and you have no idea *why* he was open. The why happened in the first second, off screen.

! Why it exists: the tight frame

Why don't broadcasts just stay wide? Because the wide shot loses more viewers than it gains. On a wide frame each player is a few pixels tall, the ball is a dot, and a running play turns into a blur of jerseys. The tight frame makes the result easy to see for everybody, at the cost of the cause. Broadcasts compensate with replays, the subject of the next section, and some now offer a second, wider feed for viewers who want it (the film room at the end of this chapter looks at one). There's a trade-off for you too: if you want the cause, you have to go looking for it, in the replays and in the All-22.

Getting the All-22 yourself

For decades All-22 film was something only teams had. That changed in 2012, when the NFL began selling fans the coaches' sideline and end-zone angles of every play through its online Game Rewind service.² Today the league's streaming service, NFL+, includes All-22 coaches film in its paid Premium tier: for completed games, every snap from the sideline and end-zone angles, with no broadcast cut-aways.³ The All-22 is where the full answer to every "why was he open?" lives, and Film Study: All-22, Charting a Game, and Scoring Yourself gives you a method for watching it. You don't need it yet. This chapter is about the broadcast, because that is what you'll actually watch on Sunday.

Replays: the second look the broadcast gives you

The live picture shows the result. The replays are where a broadcast tries to show the cause, and they come in a fairly predictable order on a big play. The producer in the truck chooses the angles while the teams are getting ready for the next snap, roughly in order of how quickly each is ready and how well it tells the story.

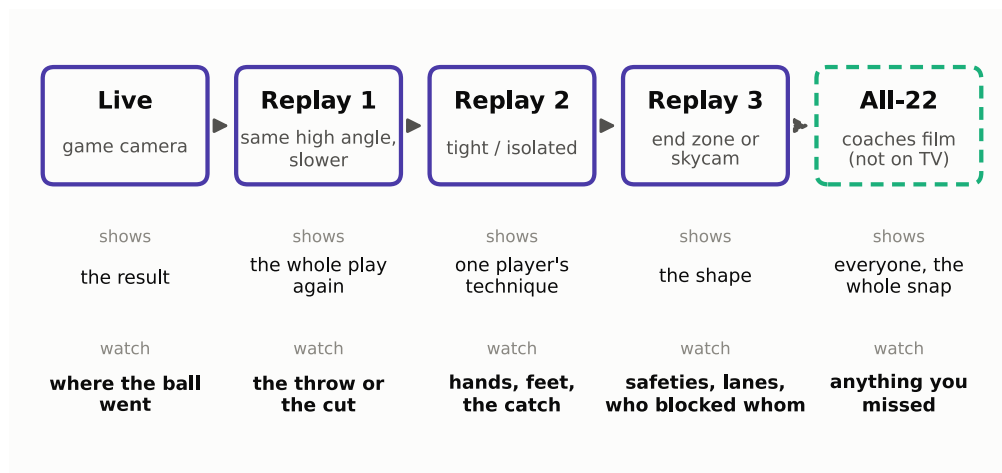


Figure 3: A typical replay sequence after a big play (the order and the angles vary with the network and the play). Each angle answers a different question, so watch something different on each one. The last box is not on TV: only coaches film shows every player for the whole snap.

The live shot and the first replay tell you *what*: the receiver caught it, the back broke a tackle. The tight replay shows *how* one player did it: the receiver's hands, the cornerback's feet, the running back's cut. The end-zone or skycam replay, usually the last

²NFL.com, "NFL.com to offer fans coaches film with NFL Game Rewind" (June 2012): <https://fantasy-www.nfl.com/news/nfl-com-to-offer-fans-coaches-film-with-nfl-game-rewind-09000d5d82ac698a> ; *SportsBusiness Daily*, June 18, 2012: <https://www.sportsbusinessdaily.com/Daily/Issues/2012/06/18/Media/NFL-22.aspx> (the "All-22" and "High End Zone" angles for every 2012 play, plus the 2011 archive).

³NFL, "NFL+: learn more" (tiers and features, including All-22 coaches film in NFL+ Premium): <https://www.nfl.com/plus/learn-more> ; NFL.com, on NFL Pro, which offers All-22 film back to the 2022 season with Next Gen Stats search: https://www.nfl.com/_amp/launch-of-nfl-pro-highlights-third-season-of-nfl

one and often cut short when the offense gets to the line, is the one to wait for, because it shows the *shape* of the play: where the safeties went, the lane the ball traveled through, which defender a blocker took. If you only have the patience for one replay, make it that one.

The habit that makes replays useful is simple: **watch something different on each one**. You saw the result live. On the first replay, ignore the ball and watch the safeties. On the second, watch the defender nearest the receiver. On the end-zone angle, watch the line. Replays are short, and a producer's sequence rarely runs more than three or four angles before the next snap, so choose before the replay starts.

The five-second pre-snap read

Most of what you can learn from a broadcast, you learn before the snap, and it starts with the score graphic in the corner of the screen. Everything in it changes what both teams are likely to do, and once you know what to look for you can read all of it in about five seconds.



Figure 4: An illustrative score graphic (team names and numbers invented; real layouts differ by network) and the five-second read it supports. Each item answers a question about what the teams need: whether a pass is likely, whether anyone needs to hurry or burn clock, and how much a failed play would cost. The yellow line on the field, the line to gain, is the sixth item.

Work through the graphic in the same order every time, because the order runs from the strongest clue to the weakest.

- 1. Down and distance.** This is the single best predictor of run or pass you will see all game. You met it in Downs, Distance, and Field Position, where the drill was to guess run or pass from this alone. Short distances lean run, long ones lean pass, and third down leans pass unless the distance is very short. The graphic gives it to you, and the yellow line on the field shows you where the line to gain is. (That computer-drawn line is younger than many fans think: it first appeared on an ESPN broadcast of a Bengals–Ravens game on September 27, 1998.⁴)
- 2. Field position.** Where is the ball? Inside the opponent's 40 or so, a field goal becomes possible, which changes fourth down and makes a sack more costly. Deep in its own territory, an offense gets cautious. Some graphics show the yard line; otherwise read it off the field.
- 3. Score and clock.** Who is ahead, by how much, and how much time is left? A team trailing by two scores late has to pass and hurry. A team protecting a lead late wants to run and keep the clock moving. In the first three quarters of a close game, neither matters much; that's what analysts call a "neutral" situation.
- 4. Timeouts.** The dots under each team's name. They matter late in a half: a team with no timeouts can't stop the clock except by throwing incomplete or getting out of bounds, which changes what it calls.

Many graphics now show the **play clock** too. A low play clock with the offense still not set tells you the call came in late, or that the quarterback is changing it (the last fifteen seconds before the snap are the subject of the previous chapter).

⁴Wikipedia, "1st & Ten (graphics system)" (developed by Sportvision; first broadcast on ESPN's Bengals–Ravens game, Sept. 27, 1998): [https://en.wikipedia.org/wiki/1st_%26_Ten_\(graphics_system\)](https://en.wikipedia.org/wiki/1st_%26_Ten_(graphics_system)); Mental Floss, "Explaining the magic yellow first-down line": <https://mentalfloss.com/article/27009/explaining-magic-yellow-first-down-line>

The point of the read is not to memorize the numbers. It is to turn them into **one sentence about what each side needs**: “third-and-6, in field-goal range, down 3: pass likely; a stop forces a kick.” That sentence is your prior: the guess you make before you’ve seen a single player line up, which everything else you notice will adjust.

The numbers on the screen

Broadcasts now put model outputs on screen alongside the score. Two families are worth knowing now, glossed in a line each, because the models behind them are taught later.

The win-probability bar. The “WP HOME 58%” in the graphic is a **win probability**: an estimate of how often teams in exactly this situation (this score, time left, field position, down and distance, timeouts) have gone on to win, from a model fit to many past games. Some models also fold in which team was favored before the game. It moves with every play, and the plays that move it most are the ones that mattered most. Two warnings. First, 58% is close to a coin flip with a slight lean, not a prediction that the home team will win; and 90% means that about one team in ten in this spot still loses. Second, different broadcasters use different models, so two networks can show different numbers for the same moment. How these models are built, and how coaches use them on fourth down, is in Win Probability and Decision Analysis.

Next Gen Stats overlays. Every NFL player wears small radio chips in his shoulder pads, and receivers around each stadium record every player’s position about ten times a second; the ball carries a chip too. The league packages that tracking data as **Next Gen Stats**.⁵ On a broadcast it shows up as numbers like these:

- **Speed** (in miles per hour): how fast the ball carrier or a chasing defender was moving at his peak.
- **Separation**: the distance, in yards, between a receiver and the nearest defender when the ball arrived (or was thrown). More separation, easier catch.
- **Completion probability**: how often a throw like this one (its distance, the receiver’s separation, the pressure on the quarterback) is completed. A catch with a 25% completion probability is a hard one.
- **Time to throw**: seconds from the snap until the ball left the quarterback’s hand. Fast times usually mean a quick, pre-planned throw; long ones mean good protection or a scramble.

Treat these as you’d treat any measurement: useful, specific, and narrower than they sound. “Completion probability 25%” says the throw was hard, not that the quarterback was brilliant; it doesn’t know whether the receiver ran the wrong route. How tracking data works, and how to load and plot it yourself, is in Tracking Data and the Big Data Bowl.

Common misconception

“The win-probability bar jumped from 30% to 70%, so that was the best play of the game.” Maybe. A swing that large does say the play changed the game’s likely outcome more than any other. But win probability measures *consequence*, not *quality*: a routine three-yard catch on fourth-and-2 in the final minute can swing it more than a brilliant 40-yard catch in the first quarter. For “how good was the play, given the situation,” the per-play scorecard from Downs, Distance, and Field Position, EPA, is the better tool.

Watch one thing

Here is the most useful change you can make to how you watch: **stop following the ball**. The camera follows it for you. Your eyes are free to watch something the camera won’t. The catch is that you can’t watch everything, because a play is over in five or six seconds and twenty-two people are moving. So pick **one thing** for each phase of the snap and watch only that.

⁵NFL Football Operations, “NFL Next Gen Stats”: <https://operations.nfl.com/gameday/technology/nfl-next-gen-stats/> (chips in each player’s shoulder pads and in the ball; positions recorded about ten times a second). The system was tested in 2014 and in use league-wide from 2015; see Arrow Electronics, “Wearable RFID is changing the nature of NFL statistics”: <https://arrow.com/en/research-and-events/articles/wearable-rfid-is-changing-the-nature-of-nfl-statistics>

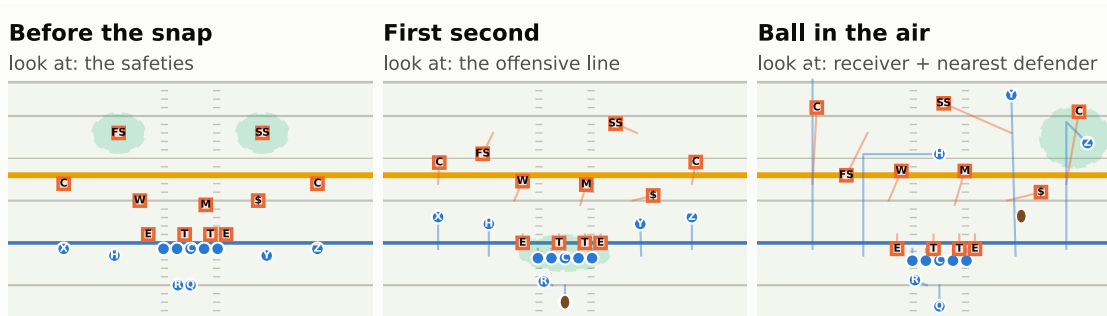


Figure 5: Where to look, by phase, on the chapter’s pass play (simulated). Before the snap, the safeties: two deep or one, and how deep. In the first second, the offensive line: helmets moving back means a pass, firing forward means a run. While the ball is in the air, the receiver and the nearest defender, then the safety arriving. The aqua areas are where your eyes should be; notice how little of it is near the ball.

Before the snap, the TV gives you a wide frame, so spend it on the defense’s deep players. How many safeties are deep, and how deep? Two safeties deep, roughly 12 to 15 yards, suggests the defense is guarding against long passes first. One deep in the middle with the other walked down toward the line suggests it wants an extra defender near the ball, against the run or to cover a receiver man to man. These pictures, called **two-high** and **one-high**, are the first defensive read in this course, and A First Look at Defense builds them properly. If you have time left, glance at the backfield: where the running back stands is a clue of its own (see the tells below).

In the first second after the snap, watch the offensive line. It is in frame, it is easy to see, and it answers the most basic question first. On a pass, the linemen’s helmets move *backward* as they set up to protect; on a run, they fire *forward* into the defenders. You will often know run or pass before the quarterback has finished turning around, and with practice you can see more: which direction the line is moving, and whether a guard is pulling around the end (running sideways along the line to lead a run, explained in The Language of the Run Game).

While the ball is in the air, watch the receiver and the defender nearest him, and then the deep safety arriving. That is where the play is decided, and the camera will show you the catch whether you look for it or not; what it won’t always show is the defender’s angle and the safety’s path, which tell you whether the coverage was beaten or simply arrived late.

What you watch depends on what you expect, so the pre-snap read feeds straight into this. A rough guide by kind of play:

If you expect...	Watch first	Why
A run between the tackles	The two guards, then the linebackers’ first step	The guards show where the run is aimed; linebackers who step up and fill the gap stop it, ones who hesitate don’t.
A run to the outside	The defender at the end of the line on that side	If he holds the edge, the runner has to cut back inside; if he’s pinned or sucked inside, the run can get outside.
A dropback pass	Safeties before the snap, then the quarterback’s head	The safeties tell you where the defense is thin; the quarterback’s eyes usually point where the ball is going.
Third-and-long	The defenders at the line to gain	The offense needs to get past them; watch whether the throw goes beyond them or in front of them.
Short yardage or goal line	The linemen’s pad level	The lower man usually wins. The pile is decided in the first half-second.
A replay	Whatever you didn’t watch live	Each replay angle is a free second look at a different group.

Gap and Power Runs, Dropback Concepts and Third Down turn each row into a full chapter. For now the habit matters more than the details: one thing per phase, chosen before the snap.

Common misconception

“If I don’t follow the ball, I’ll miss the play.” You won’t. The camera follows the ball for you, the crowd will tell you when something big happens, and the replay will show you the result again, usually within ten seconds. What you miss by following the ball is everything else, and on a live broadcast there’s no second chance to see that.

Tells: clues the players give away

The pre-snap read so far used only the graphic. The players themselves give away more. A **tell** is a pre-snap clue to what a team is about to do: a way a player stands, where he lines up, how far away he is. The word comes from poker, and it means the same thing here: a clue that shifts the odds, which the other side is working to hide or fake. **Every tell is a clue, never a guarantee.** Coaches study their own film to find and remove their tells (a habit called self-scouting), and the smartest ones show a tell on purpose and run the opposite play.

Four tells are worth learning first, because they are visible on an ordinary broadcast.

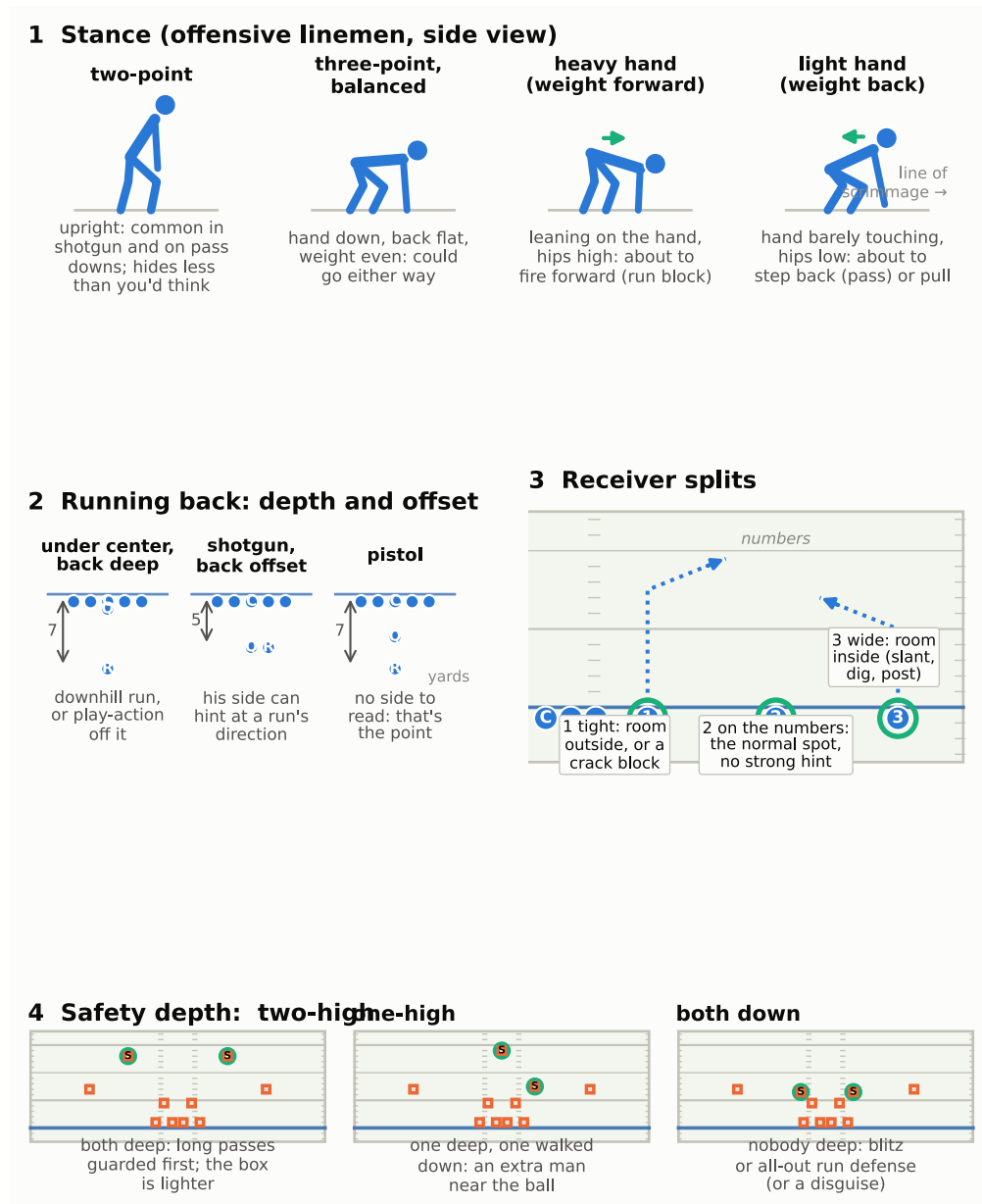


Figure 6: The starter tells card. (1) Linemen's stance, seen from the side: weight forward on the down hand hints at a run, weight back at a pass set or a pull. (2) The running back's depth and offset. (3) A receiver's split, his distance from the ball: it decides how much room he has to the inside and outside. (4) Safety depth and number. Each is a lean, not a rule, and the offense or defense may be faking it.

Stance

A **stance** is how a player stands before the snap. In a **two-point stance** he is upright with his hands off the ground (hands on his thighs or up in front of him); in a **three-point stance** he crouches with one hand on the ground, two feet and a hand making three points. Linemen are the players to watch, and the tell is where their **weight** is. A lineman about to fire forward into a run block wants his weight out over his hand, so he can explode straight ahead: hips a little high, shoulders low, the down hand bearing weight. That's a **heavy hand** (or **heavy stance**). A lineman about to step *back* into a pass set, or step sideways to pull, wants his weight on his feet, so his hand barely touches the ground and his hips sit lower: a **light hand**. On a close-up of the line before the snap you can sometimes see the difference in the fingers: a lineman leaning on his hand has white knuckles and his fingers spread and bent; a light hand looks like it's resting.⁶

⁶On reading linemen's stances and weight distribution from the stands or the broadcast, see Pat Kirwan, *Take Your Eye Off the Ball* (Triumph Books, 2010), ch. 2–3. Line coaches' own teaching of a balanced stance that hides run/pass intent is standard; how much weight shows varies by player and is a judgment call, not a measurement.

Why does a tell like this exist at all, when every line coach knows about it? Because physics gets a vote. A stance that's ideal for firing forward is a worse stance for stepping back, and vice versa. Coaches teach a single balanced stance for every play precisely to remove the tell, but a lineman facing a 300-pound defender wants every inch of advantage, and under pressure weight drifts toward the job he's about to do. The defensive linemen across from him are reading the same clue, from two feet away.

Two complications. First, many offenses now line up in two-point stances on most snaps, especially in the shotgun, where the line pass-blocks so often that standing up is simply better; then the tell is a *change*, such as a line that is usually upright putting a hand down on second-and-2. Second, offenses that know they have this tell sometimes run play-action (a fake handoff followed by a pass, built in Play-Action, Bootlegs, and Screens) out of their heaviest run stances, exactly because the defense is reading them.

The running back's depth and offset

Where the running back lines up is the second tell, and it comes in two parts: how deep and to which side.

Depth. A back seven yards deep behind a quarterback who is under center (directly behind the center, taking the ball from his hands) is set up to take a handoff running downhill, already at speed when he reaches the line. That alignment leans run, and it is also the alignment every play-action pass is built to look like. A back standing level with a quarterback in the **shotgun** (the quarterback about five yards behind the center, receiving the snap through the air) is in the best place to block for a pass or to take a quick handoff; that alignment leans pass. The 2025 numbers are striking: with the quarterback under center, offenses passed on about a third of plays; from the shotgun, on about three quarters.⁷ The quarterback's alignment is the single strongest tell on most broadcasts, and the backfield names (under center, shotgun and pistol) are defined properly in Part 2.

Offset. In the shotgun the back has to stand to one side of the quarterback, and which side can hint at where a run goes. On many common shotgun runs the back takes the handoff while crossing in front of the quarterback, so the run starts toward the side *away* from where he lined up; on others he runs to his own side. Each offense has its habits, and defenses scout them weekly, which is why this is a clue to check against the team you're watching rather than a rule. The offense's answer to this tell is a formation built to remove it: the **pistol**, in which the quarterback stands about four yards back and the running back directly behind him, so there is no side to read. That is one of the main reasons the pistol exists.

Receiver splits

A receiver's **split** is how far he lines up from the ball (or from the nearest lineman), and it matters because the sideline is a wall. A receiver who lines up close to the formation, a tight or reduced split, has lots of room to his *outside*: room to run an out or a corner route toward the sideline. He is also close enough to the line to turn and block a defender inside him, a *crack block* on a run to his side (Perimeter Runs and Misdirection). A receiver split very wide, near the sideline, has little room outside, so the routes that make sense for him go *inside*: slants, digs, posts. Defensive backs are taught exactly these "split rules."⁸ The receiver on the numbers (the painted yard-line numbers, about twelve yards from the sideline) is in the normal spot and tells you little.

Splits are the easiest tell for an offense to fake and the easiest for you to misread, because they also change with the ball's position: on the hash nearer one sideline, everyone on that side looks "wide." Use them as a second clue, not a first one.

Safety depth

The last tell is on the other side of the ball, and you have already met it: how many safeties are deep and how deep they are. Two safeties at 12 to 15 yards (two-high) mean the defense has committed two players to stopping long passes, leaving fewer players near the line, the area called the box, to stop a run. One deep safety with the other walked down to about eight yards (one-high) puts an extra defender near the ball. Both safeties down near the line suggests a blitz (more than four players rushing the quarterback) or an all-out run defense, and it leaves nobody deep to stop a long pass.

This tell has a twist: it's the one defenses most often fake. The opening play of this chapter was exactly that, two-high before the snap and one-high after it. Safeties can move a lot of ground in the second after the snap, and defensive coordinators know you're watching them. Read the safety depth before the snap, then check, on the replay, whether it held. Disguise, Rotation, and Split-Field Coverage is the long version.

⁷Course calculation from nflverse play-by-play (shotgun and pass flags), 2025 regular season, runs and passes only: passes were 32% of snaps with the quarterback under center and 75% of shotgun snaps (sacks and scrambles counted as passes; **shotgun** = 0 includes the pistol in this data). Neutral situations (outside the last two minutes of each half, win probability 20–80%): second-and-1 or 2 under center, 19% passes (370 snaps); third-and-7 to 10 from the shotgun, 97% passes (1,069 snaps).

⁸Receiver "split rules" for defensive backs (wide split: expect an inside-breaking route; tight split: expect an outside-breaking route or a crack block) are standard coaching material; see, e.g., Pat Kirwan, *Take Your Eye Off the Ball* (2010), and the defensive-back clinic literature summarized in Chris B. Brown, *The Art of Smart Football* (2015).

These four are the starter set. The full catalogue (tight end stance, linebacker depth, the cornerbacks' depth and inside or outside position, the linemen's weight on pass sets and more) is built up chapter by chapter and collected in Watching Live: The Master Checklist.

How good is a guess?

Before you start keeping score on yourself, it helps to know what a good score looks like. Guessing run or pass sounds like a coin flip, so 50% might seem like the baseline. It isn't. In the 2025 regular season about 60% of plays were passes (counting sacks and scrambles, which begin as passes), so a viewer who said "pass" on every snap would have been right 60% of the time without looking at the screen. That is the floor. The question is how much each clue adds.

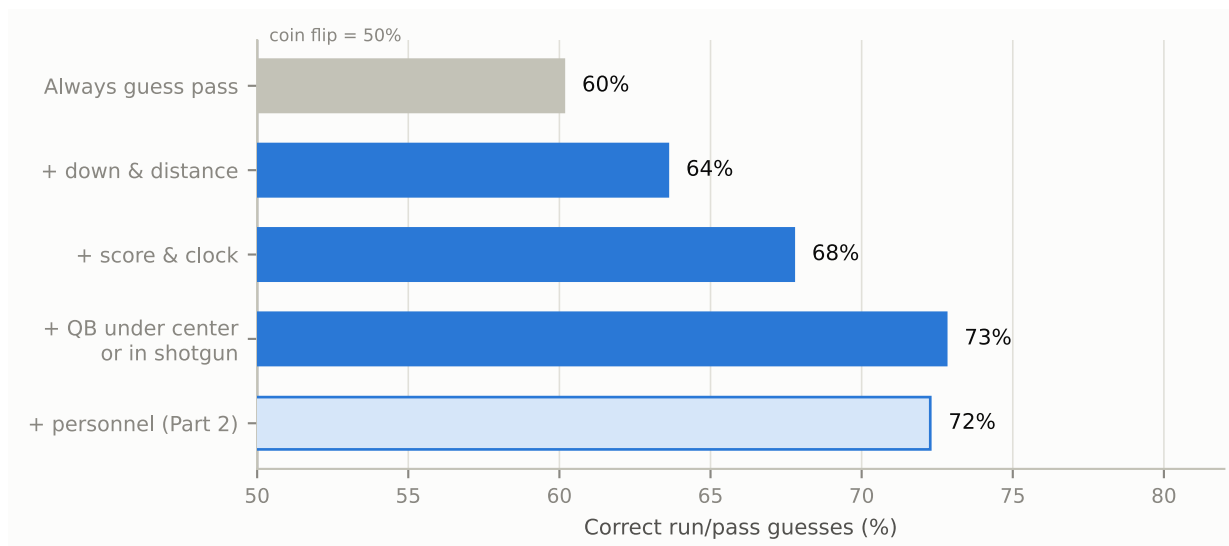


Figure 7: How often a simple rule guesses run or pass correctly, 2025 regular season. Each bar uses the clues above it plus one more and always guesses whichever (run or pass) was more common in that situation. Rules were built from half of the season's games and scored on the other half, then the halves swapped. The last bar adds a clue taught in Part 2 (personnel: how many backs, tight ends and receivers are on the field). Source: nflverse play-by-play and FTN participation data; runs and passes only, sacks and scrambles counted as passes.

Read the bars from the top. Saying "pass" every time scores about 60%. Knowing the down and distance, the first item in the five-second read, lifts a simple rule to about 64%. Adding the score and the clock gets it to about 68%, and adding one tell, whether the quarterback is under center or in the shotgun, to about 73%. That is the bar for a viewer who uses the graphic and one glance at the backfield. Personnel, the first thing Part 2 teaches, adds a little more (about 72%), and the rest of the course adds clues these rules can't see: formation, motion, the defense's alignment, a team's habits.

Two things follow. First, **your raw accuracy means little on its own**. If you hit 65% in a game where the offense threw on three of every four snaps, you did worse than "always pass." Always compare yourself to the baseline for the game you watched. Second, the gains get harder. Every clue after down and distance adds a few points, not twenty. A viewer who hits 75% over a full game is reading the field well; 85% over many games would be exceptional, because offenses work hard to be unpredictable, and the best play-callers are deliberately difficult to guess.

The prediction log

Which brings us to the habit the rest of the course is built on. In Downs, Distance, and Field Position you guessed run or pass from down and distance and kept a tally: that was your baseline. From now on you keep a **prediction log**: a running record of your pre-snap guesses, the clue you used, and what happened. It starts simple, run or pass, and grows as the course does: by Part 2 you'll log the personnel and formation, by Part 6 the coverage you expect, and in the capstone you'll attach a confidence to each guess and score the whole season's log properly.

PREDICTION LOG									
Game: _____ at _____ Date: _____ Half: 1st Team on offense: both									
#	Qtr	Clock	Down & dist	Ball on	Score	Clue I used	Guess	Result	Hit
1	1	14:55	1st & 10	own 25	0-0	gun, back offset	P	R	X
2	1	14:20	2nd & 6	own 29	0-0	under center, back deep	R	R	✓
3	1	13:41	3rd & 2	own 33	0-0	heavy hands, tight splits	R	R	✓
4	1	13:02	1st & 10	own 36	0-0	gun, 2 safeties deep	P	P	✓
5	1	12:27	2nd & 10	own 36	0-0	gun, long distance	P	P	✓
6	1	12:21	3rd & 10	own 36	0-0	3rd & long	P	P	✓

Hits: ____ / ____ = ____ % 'Always the more common one' (count P and R in the Result column): ____ % Beat it by: ____

Figure 8: A prediction log on paper (the first rows are an invented example). One row per snap. Fill in the situation and your guess before the snap, the result after it. The ‘clue’ column is the one that teaches you: when a guess is wrong, it tells you which clue misled you.

The paper version is the real one, because you can fill it in during a commercial break with no screen in your hand. Five rules make it honest:

- Write the guess before the snap.** Not after the handoff, not “I knew it.” If you missed the snap, leave the row blank.
- Write the clue.** One or two words: “3rd & long,” “under center, back deep,” “two high.” Over a season, the clue column tells you which clues you trust and how often they were right.
- Count every snap you guessed,** easy ones included. A log of only the hard calls hides how much the easy ones flatter you, and a log of only the easy ones flatters you more.
- Log the play as it began, not as it ended.** A pass play where the quarterback scrambles is a pass; a fake handoff followed by a throw is a pass. The question is what the offense called.
- Score against the baseline,** not against 50%: count how many plays were passes, and see whether you beat “always pass” (or “always run,” in the rare game where runs win).

If you’d rather type than write, a spreadsheet with the same columns works, and a few lines of Python do the scoring. Save your log as a CSV with columns qtr, down, dist, clue, guess, actual (guess and actual are R or P), then:

```
import pandas as pd

log = pd.read_csv(LOG_FILE) # e.g. LOG_FILE = "my_log.csv": one row per snap
log["hit"] = log["guess"] == log["actual"]
share_pass = (log["actual"] == "P").mean()
print(f"My accuracy: {log.hit.mean():.0%} on {len(log)} snaps")
print(f"'Always the common one': {max(share_pass, 1 - share_pass):.0%}")
print(log.groupby("down").hit.agg(["mean", "size"]).round(2)) # where am I good?
print(log.loc[~log.hit, ["qtr", "down", "dist", "clue", "guess", "actual"]]) # what fooled me?
```

```
My accuracy:      44% on 34 snaps
'Always the common one': 56%
      mean size
down
```

1	0.23	13					
2	0.62	13					
3	0.50	8					
	qtr	down	dist		clue	guess	actual
3	1	1	10	under center, long	P	R	
4	1	2	2	under center	R	P	
8	1	1	10	under center, long	P	R	
9	1	2	6	under center	R	P	
10	1	3	3	gun	P	R	
11	1	1	10	gun, long	P	R	
13	1	1	10	gun, long	P	R	
15	1	3	5	under center	R	P	
16	1	1	10	gun, long	P	R	
17	2	2	8	under center, long	P	R	
19	2	1	10	gun, long	P	R	
20	2	2	2	under center	P	R	
21	2	1	10	under center, long	P	R	
24	2	1	10	under center, long	P	R	
26	2	3	1	gun	P	R	
28	2	2	6	gun	P	R	
29	2	1	10	under center, long	P	R	
31	2	3	12	under center, long	R	P	
32	2	1	10	gun, long	P	R	

On the invented sample half above, the viewer beat “always pass” by a few points, was nearly perfect on third down, and was fooled mostly on first down from under center: a play-action offense, or a viewer leaning too hard on one clue. That last table, the misses with their clues, is where you learn. In the capstone, Film Study: All-22, Charting a Game, and Scoring Yourself adds confidences to each guess and scores the season’s log with proper tools.

Film room

i Film room: Patriots 2014 — the replay that explained Super Bowl XLIX

Super Bowl XLIX, February 1, 2015. Seattle trails New England 28–24 and has the ball second-and-goal at the Patriots’ 1-yard line with 26 seconds left. Russell Wilson takes a shotgun snap and throws a quick slant to Ricardo Lockette, and Malcolm Butler, an undrafted rookie cornerback, arrives at the same moment as the ball and intercepts it. The Patriots win.⁹

What the live picture showed: a pile of bodies at the goal line and a ball that ended up in the wrong hands. From the high sideline angle, tight on the quarterback at the snap, the play looked like chaos.

What the replays showed, especially the angle from behind the end zone: two Seattle receivers lined up one behind the other on the same side (a stack). The front one, Jermaine Kearse, was supposed to drive into the defenders and screen them off so the back one, Lockette, could cut inside underneath him: a rub route, legal as long as the receiver doesn’t deliberately block before the ball is thrown. Cornerback Brandon Browner jammed Kearse at the line, so the screen never happened, and Butler, unbothered, broke on the slant before Lockette had turned. Butler later said the Patriots had practiced against this look and he recognized it.¹⁰

What to notice: the answer was in the alignment and in two defenders’ first steps, the two things a ball-following camera shows worst. A stack at the goal line is a tell in its own right. The end-zone replay is the angle that makes a play like this make sense, which is why it’s the one to wait for.

⁹NFL.com, “NFL 100 greatest plays: Malcolm Butler interception in Super Bowl XLIX”: <https://nfl.com/100/originals/100-greatest/plays-5> ; NFL.com, “Butler’s late INT lifts Patriots to fourth Super Bowl title”: https://nfl.com/_amp/butler-s-late-int-lifts-patriots-to-fourth-super-bowl-title-0ap3000000467455 (score, 26 seconds, second-and-goal from the 1, Browner’s jam on Kearse).

¹⁰NBC Sports Boston, “Malcolm Butler gives great insight into Super Bowl XLIX interception”: <https://www.nbcsportsboston.com/nfl/new-england-patriots/malcolm-butler-gives-great-insight-into-super-bowl-xlix-interception/385317/>

i Film room: Prime Video 2022–2025 — a broadcast that reads tells for you

When Amazon’s Prime Video took over *Thursday Night Football* in 2022, it added an alternate stream, *Prime Vision with Next Gen Stats*, built for viewers who want more than the ball. During plays it switches to a higher, wider camera (much closer to All-22 than a normal broadcast), it traces receivers’ routes live, and it fills the edges of the screen with data.¹¹ In 2023 Prime Video added **Defensive Alerts**: before the snap, a model trained on tracking data circles in red the defenders it judges most likely to rush the quarterback.¹² It later appeared on the main broadcast as well.

What to notice: Defensive Alerts is doing what this chapter asks you to do. It reads pre-snap clues (a defender’s depth, his alignment, his movement in the last seconds before the snap) and turns them into a probability, not a certainty. When the red circle is wrong, ask what the defender did to fool it: often that is a disguise, which is to say a tell faked on purpose. If you get the chance, watch a quarter on the alternate stream and a quarter on the main one; the difference in how much of the defense you can see is the argument of this chapter in one sitting.

🔗 Watch for it on Sunday: prediction log, game 1

For **one half** of one game, keep the log. Paper, a pen, and the template above.

- **Before each snap**, do the five-second read (down and distance, field position, score and clock, timeouts), take one look at the backfield (under center or shotgun? where’s the back?), and write **R** or **P** with your clue. Use the commercial breaks to catch up.
- **During the snap**, watch one thing: the safeties before the snap, the offensive line in the first second. Don’t follow the ball; the camera will.
- **After the snap**, write what the offense ran. Use the replay if you’re not sure.
- **At halftime**, score it: your hits, the share of passes in your log, and the gap between them. Then compare with your down-and-distance baseline tally. Did the backfield glance help?
- **Keep the sheet.** This is page one of a log you’ll extend in every Part of the course.

¹¹Sportico, “Amazon’s Prime Vision Thursday Night Football feed” (2022): <https://www.sportico.com/business/media/2022/amazon-prime-vision-thursday-night-football-feed-1234691348/> ; Sports Video Group, “Amazon’s Jared Stacy on iterative change in season 2 of Thursday Night Football on Prime” (Sept. 14, 2023): <https://www.sportsvideo.org/2023/09/14/svg-sit-down-amazons-jared-stacy-on-iterative-change-in-season-2-of-thursday-night-football-on-prime/>

¹²Amazon, “Prime Video’s Thursday Night Football: Next Gen Stats AI features” (Defensive Alerts, Prime Targets, Pressure Probability and others): <https://aboutamazon.com/news/aws/prime-video-thursday-night-football-next-gen-stats-ai-features>

Predict the play

💡 Predict the play 1: second-and-1

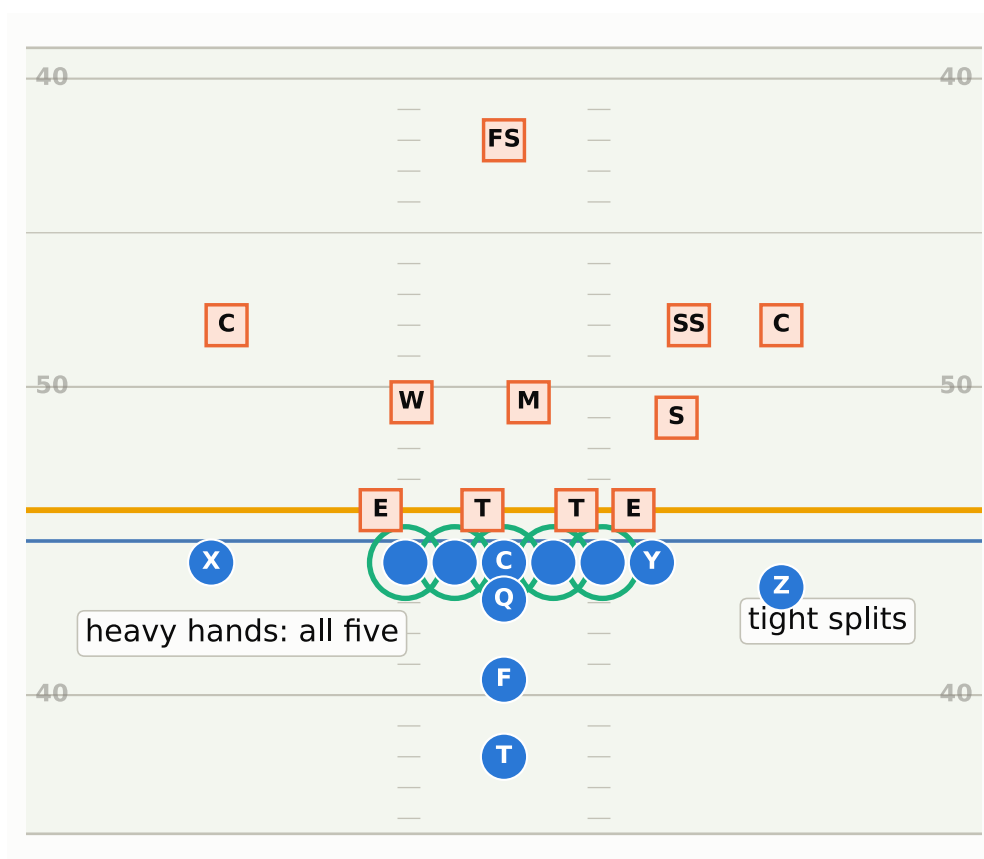


Figure 9: Second-and-1 near midfield, early in the second quarter, tied 7-7. The quarterback is under center with a fullback (F) and a tailback (T) lined up behind him, the tight end beside the right tackle, and both wide receivers in tight splits. On the close-up before the snap, all five linemen have a hand on the ground and are leaning on it. The defense has one safety deep and the other walked down.

Run or pass? And what's the risk in your guess?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: third-and-8

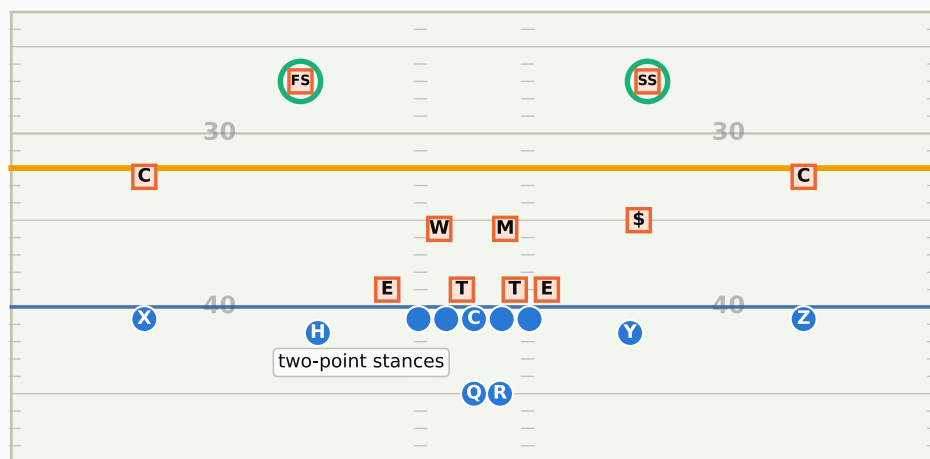


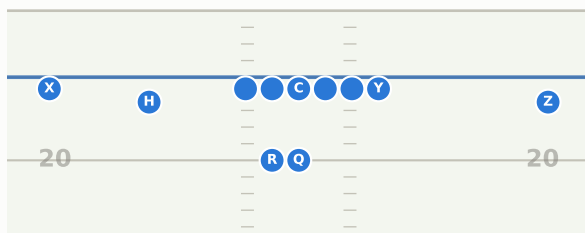
Figure 10: Third-and-8 at the opponent's 40, third quarter, home team down 4. The quarterback is in the shotgun with the back beside him. The linemen are standing in two-point stances, and the outside receivers are split wide. Two safeties stand about 13 yards deep.

Run or pass? And where should your eyes go in the first second, given your answer?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: which backfield tells you more?

A shotgun, back offset



B quarterback 4 yards deep, back behind

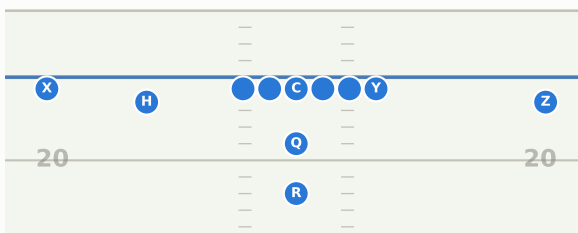


Figure 11: Two backfields on first-and-10, both from the same offense. In A, the quarterback is in the shotgun and the back stands to his left. In B, the quarterback is about four yards deep and the back directly behind him.

If the offense runs, which picture gives you a better guess at the direction, and why would an offense choose the other one?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **The TV picture follows the ball**, so on most pass plays it hides the safeties, the cornerbacks and most receivers after the snap. The cause of a big play usually happens off screen in the first second.
- **Use the replays deliberately**: watch something different on each one, and wait for the end-zone or skycam angle, which shows the shape of the play. All-22 coaches film shows everything, every snap.
- **Read the graphic in five seconds** (down and distance, field position, score and clock, timeouts) and turn it into one sentence about what each side needs. Win probability measures a situation's consequences, not a play's quality; Next Gen Stats numbers are measurements with narrow meanings.

- **Watch one thing per phase:** the safeties before the snap, the offensive line in the first second, the receiver and nearest defender while the ball is in the air. Let the camera follow the ball.
- **Tells are clues, not guarantees:** stance, the back's depth and offset, receiver splits and safety depth each shift the odds, and each can be faked. The quarterback's alignment alone moves a simple run/pass rule from about 68% to about 73%.
- **Keep the prediction log** and always score it against the game's baseline. Your log, not your memory, is how you'll know you're getting better.

Connections

Builds on: Downs, Distance, and Field Position (the down-and-distance baseline this log starts from), What a Play Really Is (one call is a family of outcomes, which is why reading the first second matters) and The Pre-Snap Phase (the movement you'll watch for in the last seconds before the snap).

Sets up: Part 2 gives your log its next clues: Personnel (who's on the field), Formation Rules and the Language of Alignment (splits, backfields, the pistol) and A First Look at Defense (the box, one-high and two-high). Further on, Win Probability and Decision Analysis builds the model behind the on-screen bar, Tracking Data and the Big Data Bowl the data behind Next Gen Stats, Watching Live: The Master Checklist completes the tells catalogue, and Film Study: All-22, Charting a Game, and Scoring Yourself teaches All-22 study and scores the log you start today.

Go deeper

- Pat Kirwan, *Take Your Eye Off the Ball* (2010; 2nd ed. 2015): the classic book on watching football away from the ball, by a former NFL coach and scout. Its chapters on the pre-snap picture are the natural companion to this one.
- Chris B. Brown, *The Art of Smart Football* (2015): short essays that make the “why” behind what you're seeing on the All-22 clear.
- NFL Next Gen Stats (nextgenstats.nfl.com): the league's own tracking-data numbers, with definitions of each stat.
- NFL+ Premium for All-22 coaches film; when you're ready to study a game properly, start there.

Answers to Predict the play

Answer 1.

Run, fairly confidently. Every clue points the same way. Second-and-1 is a run down for most teams; in neutral situations in 2025, offenses with the quarterback under center ran on roughly four of five second-and-1-or-2 snaps. Two backs stacked behind the quarterback are set up to run downhill. Tight splits put the receivers close enough to block, and five heavy hands say the line is about to fire forward. The defense agrees: the safety walked down makes eight defenders near the line.

The risk is play-action. This is exactly the picture a play-action pass is designed to copy, and second-and-1 is a favorite down for it, because the offense can afford to fail: an incompletion leaves third-and-1. A defense loaded up to stop the run has one deep safety, and if the linebackers and that walked-down safety bite on the fake, there's a lot of open field behind them. So your log entry is “R,” but your eyes, in the first second, should go to the line: if the linemen fire forward, it's a run; if they fire forward but stay at the line of scrimmage rather than driving into the linebackers, that's how many play-action passes look, and the ball is going over the top. (Play-Action, Bootlegs, and Screens explains why linemen can't go far downfield on a pass.)

Answer 2.

Pass, almost certainly. Third-and-8 from the shotgun is about as pass-heavy as football gets: in neutral 2025 situations, offenses facing third-and-7 to 10 from the shotgun dropped back to pass about 97% of the time. The two-point stances and wide splits agree, and the defense agrees too, with two safeties deep to keep everything in front of them.

Because the run-pass guess is easy here, it isn't the interesting prediction, and your “one thing” should move on. The interesting question is whether the throw goes **past the line to gain** or short of it. So in the first second, ignore the line and watch the two safeties and the defenders standing around eight yards deep: do the safeties stay deep (making the defense give up throws underneath and rally to tackle short of the sticks), or does one rotate down (leaving a deep area open)? The occasional run you'll see here is a **draw**, a delayed handoff designed to catch a defense that has spread out to stop the pass; if the linemen set up as if to pass and the quarterback then hands off, that's what you're watching.

Answer 3.

A tells you more. In the shotgun the back has to stand on one side, and on many common shotgun runs he takes the handoff crossing in front of the quarterback, so the run starts toward the side *away* from him: here, to the offense's right. That's a lean, not a rule; some runs go to his own side, and good offenses mix the two so that the alignment stops being a clue. Check it against the team you're watching: after a quarter, your log's clue column will tell you whether this offense has the habit.

B is the pistol, and hiding that clue is one of the main reasons it exists. With the back directly behind the quarterback, he can run to either side with the same footwork, and the defense can't lean either way before the snap. The offense also gets the downhill angle of a back lined up deep. What it gives up is a little: the quarterback is closer to the line than in the shotgun, and the back is not in his ideal spot to pass-block on either side. The pistol and its history are in Formation Rules and the Language of Alignment.